2020 Test 1 solution

Nanyang Technological University
SPMS/Division of Mathematical Sciences

AY20/21 MH2500 PROBABILITY AND INTRODUCTION TO STATISTICS Test 1

Name & Matriculation Number	Suggested Solutions	Score:	

Venue: TIME: 9:35 -10:15, 15 September, 2020

1. (40 Points) A factory finds that, on average, 20% of the bolts produced by a given machine will be defective for certain specified requirements. If 10 bolts are selected at random from the day's production of this machine, find the probability

   (a) that exactly 2 will be defective;

   (b) that 2 or more will be defective.

(a) Let X be the r.v. denoting the number of bolts selected that are defective.

$X \sim Bin(10, 0.2)$

$$P(X=2) = \binom{10}{2}(0.2)^2(1-0.2)^8$$
$$\approx 0.302$$

(b)
$$P(X \geq 2) = 1 - P(X \leq 1)$$
$$= 1 - [P(X=0) + P(X=1)]$$
$$= 1 - \left[\binom{10}{0}(0.2)^0(1-0.2)^{10} + \binom{10}{1}(0.2)^1(1-0.2)^9\right]$$
$$= 1 - 0.3758096$$
$$\approx 0.624$$

2. (30 Points) A laboratory blood test is 95% effective in detecting a ceratin disease when it is, in fact, present. However, the test also yields a "false positive" result for 1% of the healthy persons tested. (That is, if a healthy person is tested, then, with probability 0.01, the test result will imply that he or she has the disease.) If 0.5% of the population actually has the disease, what is the probability that a person has the disease that the test result in positive?

given

Let D be the event that a person has a certain disease.

Let T be the event that the test shows a positive result.

$$P(D|T) = \frac{P(T|D)P(D)}{P(T|D)P(D) + P(T|D')P(D')}$$ (Bayes' Rule)

$P(T|D) = 0.95$, $P(T|D') = 0.01$, $P(D) = 0.005$

$$\therefore P(D|T) = \frac{(0.95)(0.005)}{(0.95)(0.005) + (0.01)(1-0.005)}$$

$$\approx 0.323$$

3. (30 Points) The number of eggs laid on a tree leaf by an insect of a certain type is a Poisson random variable with parameter λ. However, such a random variable can be observed only if it is positive, since if it is 0, then we cannot know that such an insect was on the leaf. If we let Y denote the observed number of eggs, then

$$P\{Y = i\} = P\{X = i | X > 0\}$$

where X is Poisson with parameter λ. Find $\mathbb{E}(Y)$.

$X \sim \text{Poisson}(\lambda)$

$$P\{Y=i\} = P\{X=i \mid X>0\}$$

$$= \frac{P(\{X=i\} \cap \{X>0\})}{P(X>0)} \quad \text{(conditional probability)}$$

$$= \frac{P\{X=i\}}{1-P(X=0)} \rightarrow \text{complement}$$

common intersection.

$$= \frac{e^{-\lambda}\cdot\frac{\lambda^i}{i!}}{1-e^{-\lambda}}$$

$$E(Y) = \sum_{i=1}^{\infty} i\, P\{Y=i\} \quad (\text{As } Y>0)$$

$$= \sum_{i=1}^{\infty} i \left[\frac{e^{-\lambda}\cdot\frac{\lambda^i}{i!}}{1-e^{-\lambda}}\right]$$

$$= \frac{1}{1-e^{-\lambda}} \sum_{i=1}^{\infty} i\cdot\frac{\lambda^i}{i!} \quad Y>0$$

$$= \frac{1}{1-e^{-\lambda}} \sum_{i=1}^{\infty} \frac{\lambda^i e^{-\lambda}}{(i-1)!}$$

Let $j=i-1$, $\rightarrow i=j+1$

then $i=1$, $j=0$

$$= \frac{1}{1-e^{-\lambda}} \sum_{j=0}^{\infty} \frac{\lambda^{j+1} e^{-\lambda}}{j!}$$

$$= \frac{\lambda}{1-e^{-\lambda}} \underbrace{\boxed{\sum_{j=0}^{\infty} \frac{\lambda^j}{j!} e^{-\lambda}}}_{=1 \text{ (sum of Poisson pmf)}}$$

$$= \boxed{\frac{\lambda}{1-e^{-\lambda}}}$$

0-truncated Poisson Variable.